HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE413: Computer Graphics

Batch:24th		Semester: Spring 2026
Date:10/06/2026	**Midterm Examination**	Time: 1 Hour 30 Minutes
		Total Marks: 30

Answer any **five** questions. $5 \times 6 = 30$

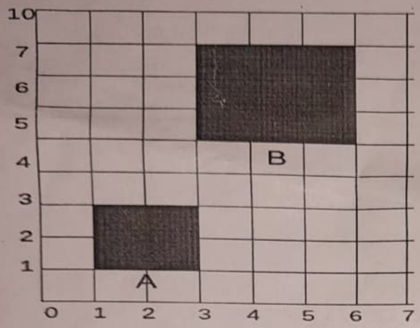
1.	(a)	Describe the operation principles of a cathode-ray tube (CRT) monitor and explain how it generates an image on the phosphor screen.	CLO1	3
	(b)	Contrast raster-scan displays with random-scan (vector) displays in terms of scanning mechanism, typical applications, and key advantages and disadvantages.	CLO1	3
2.	(a)	Explain the main idea behind the Digital Differential Analyzer (DDA) algorithm and why it is considered an incremental approach to line drawing.	CLO2	3
	(b)	Calculate the intermediate points for the line from (1,1) to (9,6) using DDA algorithm.	CLO2	3
3.	(a)	Write the pseudocode for Bresenham's line drawing algorithm.	CLO2	3
	(b)	Calculate the intermediate points for the line from (1,4) to (12,6) using Bresenham algorithm.	CLO2	3
4.	(a)	What are the differences between Bresenham's and DDA line algorithm?	CLO2	3
	(b)	What is pivot point rotation? Derive an equation for pivot point rotation.	CLO2	3
5.		For the Bresenham line drawing algorithm, derive the initial decision parameter and the equation for incremental update of the decision parameter for a line with $\Delta x > \Delta y > 0$.	CLO2	6
6.		A triangle has vertices A(2,2), B(6,2), C(4,6), apply the following transformations in sequence and calculate the final vertices i. Rotate 30° with respect to the point (2,2) ii. Translate by (5,-2)	CLO2	6

$-1 \times \sqrt{3}$

0.73

2.732

7. An object is translated from point A to B. Determine the sequence of transformations applied to the object. Justify your answer. CLO2 6



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 419: Communication Engineering

Batch : 24th
Date: 13/06/2026

Mid-Term Examination

Semester: Spring 2026
Time:1 Hour 30 Minutes
Total Marks : 30

Answer any **five** questions: $5 \times 6 = 30$

		CLOs	Marks
1.	(a) Draw and explain an ARP packet.	CLO1	3
	(b) What are the differences among Unicast, Multicast and Broadcast?	CLO1	3
2.	(a) Illustrate Cyclic Redundancy Check (CRC) encoder and decoder.	CLO4	3
	(b) State the procedure to calculate the traditional checksum.	CLO4	3
3.	(a) Explain Chunk Interleaving in forward error correction (FEC) techniques.	CLO4	3
	(b) How does a single-bit error differ from a burst error?	CLO4	3
4.	(a) Explain High-level Data Link Control (HDLC) Protocols.	CLO1	3
	(b) Write the differences between Password Authentication Protocol (PAP) and Challenge Handshake Authentication Protocol (CHAP).	CLO1	3
5.	(a) Compare and contrast byte-stuffing and bit-stuffing.	CLO1	3
	(b) Define all types of packets used in Link Control Protocol.	CLO1	3
6.	(a) Draw the taxonomy of multiple-access protocol. Explain controlled-access protocols.	CLO1	3
	(b) Describe the behavior of three persistence methods.	CLO1	3
7.	(a) Illustrate the flow diagram of CSMA/CA.	CLO1	3
	(b) Explain why collision is an issue in random access protocols but not in channelization protocols.	CLO1	3

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of CSE

CSE 469: Computer and Network Security

Mid Term Examination

Batch: 24th
Date: 20/06/2026

Semester: Spring 2026
Time: 1 hour 30 minutes
Total Marks: 30

Answer any **five** of the following questions: $5 \times 6 = 30$

1	(a)	Differentiate between cryptography and steganography with suitable examples.	CLO1	2
	(b)	Explain the CIA triad and discuss its importance in information security.	CLO1	2
	(c)	Describe the working principle and purpose of SHA in cryptographic hashing.	CLO1	2
2	(a)	Explain the basic point operations used in Elliptic Curve Cryptography, including point addition and point doubling.	CLO2	2
	(b)	Identify and explain four security mechanisms used to protect information systems.	CLO3	2
	(c)	Illustrate the key life cycle and explain the major stages involved in cryptographic key management.	CLO2	2
3	(a)	Apply Euler's theorem to compute 5^{100} mod 7.	CLO3	3
	(b)	Determine the multiplicative inverse of 8 modulo 13 using the Extended Euclidean Algorithm.	CLO3	3
4	(a)	Decrypt the ciphertext "MJQQT" using the Caesar cipher with key (K = 5).	CLO1	3
	(b)	Encrypt the plaintext "MATH" using the Affine cipher with key pair (5, 8).	CLO1	3
5	(a)	Apply the Hill cipher technique to encrypt the word "HELP" using the key matrix $k = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$.	CLO2	3
	(b)	Calculate Alice's public key (A), Bob's public key (B), and their shared secret key (S) in the Diffie–Hellman key exchange given the following parameters:Shared prime (p) = 23, Generator (g) = 5, Alice's private key (a) = 6, and Bob's private key (b) = 15.	CLO2	3
6	(a)	Analyze the limitations of the Diffie–Hellman key exchange protocol in practical security systems.	CLO2	3
	(b)	Verify whether the point P = (5, 1) lies on the elliptic curve E: $y2 \equiv x^3 + 2x + 2 \pmod{17}$.	CLO3	3
7	(a)	Compare and contrast MAC and digital signature in terms of authentication, integrity, and non-repudiation.	CLO3	3
	(b)	Verify the validity of an RSA digital signature given the following parameters: Public modulus (n) = 3233, Public exponent (e) = 17, Message hash (h) = 65, and Received signature (S) = 588.	CLO3	3

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 417: Compiler Design

Batch: 24^{th}

Date: 17.06.26

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions. $5 \times 6 = 30$

			CLO	Marks
1.	(a)	What are Alphabets and Strings? Take an example of a String and write the length of the String.	CLO2	2
	(b)	What is ambiguous grammar? Draw an NFA from the Regular Expression: (a+ U bba) + bab.	CLO4	4
2.	(a)	What is DFA? Let $\Sigma = \{a, b\}$ and let $L = \{baa\}$, design a DFA for L.	CLO4	3
	(b)	Convert DFA from NFA: L = {3rd symbol from RHS is 'b'}.	CLO4	3
3.		Derive a DFA with its transition table for the following expression: (i) Start with abaa (ii) End with bba (iii) abaabba be a substring of any string	CLO4	3*2=6
4.	(a)	Briefly explain the major phases of a compiler.	CLO1	3
	(b)	Draw the diagram of a Language processing system.	CLO2	3
5.	(a)	Write down the Role of a Parser with a diagram.	CLO1	2
	(b)	$S \rightarrow AB$ $A \rightarrow c/aA$ $B \rightarrow d/bB$ From the above grammar, draw the parse tree to derive the string: acbd.	CLO2	4

From the following code, prepare a symbol table

```
int x =10;
float y = 20.5;
void display(){
        int z = 5;
}
```

			CLO	Marks
6.	(a)	Write down the difference between Compiler and Interpreter.	CLO1	3
	(b)	Take an expression and show how the Syntax analyzer and the Intermediate code generator work on it.	CLO2	3

7. (a) $E \rightarrow TE'$ CLO4 3

$E' \rightarrow *TE' \mid \epsilon$

$T \rightarrow FT'$

$T' \rightarrow \epsilon \mid +FT'$

$F \rightarrow id \mid (E$

Find out FIRST() and FOLLOW() from this grammar.

(b) Explain the working procedure of the Semantic Analyzer. CLO2 3

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 411: Artificial Intelligence

Batch: 24^{th} — Semester: Spring 2026

Time: 1 hour 30 minutes

Date: 08/06/2026 — **Mid-Term Examination** — Total Marks: 30

Answer any **five** questions from the followings. — 5x6= 30

1. (a) Define Artificial Intelligence (AI). Classify AI and write its applications. — CLO 1, CLO 2 — 2

   (b) Given that: $f(x) = \beta_0 + \beta_1 x + e$, Write a python function that computes the $f(x)$. — CLO 4 — 2

   (c) Given that, circle is defined as a function of x, y and described as $r^2 = x^2 + y^2$. Write a python function, which computes the area of circle. — CLO 4 — 2

2. (a) Distinguish between machine learning and deep learning. — CLO 1 — 2

   (b) Given that, $v = [3,4,5]$

   i. Compute: $|v|, \hat{v}$ — CLO 3 — 2

   ii. Compute: angles with axis and corresponding relationship. — CLO 3 — 2

3. (a) Compare Supervised and Unsupervised Learning. — CLO 1 — 2

   (b) Given that: $a = [5,6,7]$ and $b = [9,0,9]$, find the relationship between a and b — CLO 3 — 2

   (c) Given that: $X = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, Compute the Determinant. — CLO 3 — 2

4. (a) Given that: $f(x) = 5x^2 + 5x + 5$ and $f(y) = 5y^2 - 5y - 5$

   i. $f'(x) - f'(y) = ?$ — CLO 3 — 1

   ii. $\int f(x)dx + \int f(y)dy = ?$ — CLO 3 — 1

   (b) Given that: $s = [0,0,0,2,2,3,5,5,5,3,4,4]$,

   i. Compute Mean, mode, median and standard deviation. — CLO 3 — 2

   ii. Write python functions for calculating mode and median. — CLO 4 — 2

5. (a) Given that, — CLO 4 — 3

Day	Sky	Rain
1	Clear	No
2	Clear	Yes
3	Cloudy	Yes
4	Cloudy	No
5	Cloudy	Yes

$P(Rain|Cloudy) = ?$

   (b) Given that,

Student	Gender	CGPA
1	Male	3.5
2	Female	2.5
3	Male	3
5	Female	3.7
6	Female	3.2

i. Find the range of Gender and CGPA. CLO 4 1

ii. Find the Q2 and Q3 for CGPA. CLO 4 1

iii. What is the average result of female student? CLO 4 1

6. Given that:

Student	Gender	Result
1	Male	Pass
2	Female	Pass
3	Transgender	Fail
5	Female	Fail
6	Male	Pass

(a) Is gender related to Result? Prove using chi-square test. CLO 4 4

Given that, chi-square distribution table:

df	P = 0.05
1	3.841
2	5.991
3	7.815

(b) Given that, mode = 5, mean = 4.5 and median = 6. Which one describe the better central tendency? Explain it. CLO 2 2

7. (a) Write a python function which computes the dot product of vector. CLO 3 2

(b) Write a python function, which solve the quadratic equation, given that, CLO 3 2

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

(c) Write a python function, which compute the addition of two given vector. CLO 3 2

4. (a) What is aliasing? Explain the concept of aliasing in signal processing and provide real-life examples where aliasing can occur. CLO2 2

(b) Consider the following analog signals: CLO2 4

$x_1(t) = \cos(2\pi \times 10t)$

$x_2(t) = \cos(2\pi \times 50t)$

Both signals are sampled at a sampling frequency:

$F_s = 40$ Hz

Determine the corresponding discrete-time signals $x_1[n]$ and $x_2[n]$. Show all mathematical steps and comment on whether aliasing occurs. Justify your answer using the sampling theorem and discrete-time frequency analysis.

5. (a) What is Fourier Transform? Explain its significance in signal analysis. CLO2 2

(b) Find the Fourier Transform of the function CLO2 4

$$f(x) = \{1, \text{ for } |x| < 1$$
$$0, \text{ for } |x| > 1\}$$

Hence, evaluate

$$\int_0^{\infty} \frac{\sin x}{x} dx$$

6. (a) Find the Fourier Transform of the function CLO2 3

$$f(x) = \{1 - x^2, |x| \leq 1$$
$$0, |x| > 1\}$$

(b) With the help of a block diagram, explain the process of Analog-to-Digital (A/D) conversion. Discuss the functions of Sampling, Quantization, and Coding, and describe how an analog signal is converted into a digital signal. CLO1 3

7. (a) What is oversampling? What are the basic elements of an oversampling D/A converter? CLO2 2

(b) With appropriate figures and examples, explain the operation of a Sample and Hold D/A converter. CLO2 2

(c) With necessary figure, briefly explain the three major steps of Fast Fourier Transform (FFT). CLO3 2